

Customer Refund Process Redesign

Reduced Turnaround Time by 65% | End-to-End Operational Improvement & System Integration Case Study

Self-directed Business Analyst portfolio project · Fictional mid-size D2C e-commerce company (~500 employees), built to demonstrate core BA deliverables

Executive Summary: Reduced average customer refund turnaround time from 12–15 business days to under 5 business days (a 65%+ reduction). Replaced 5 fragmented, email-driven cross-department handoffs with a unified case management workflow, enforced SLAs, and system integrations — projected to deliver full investment payback in 5–8 months.

1. Problem Context

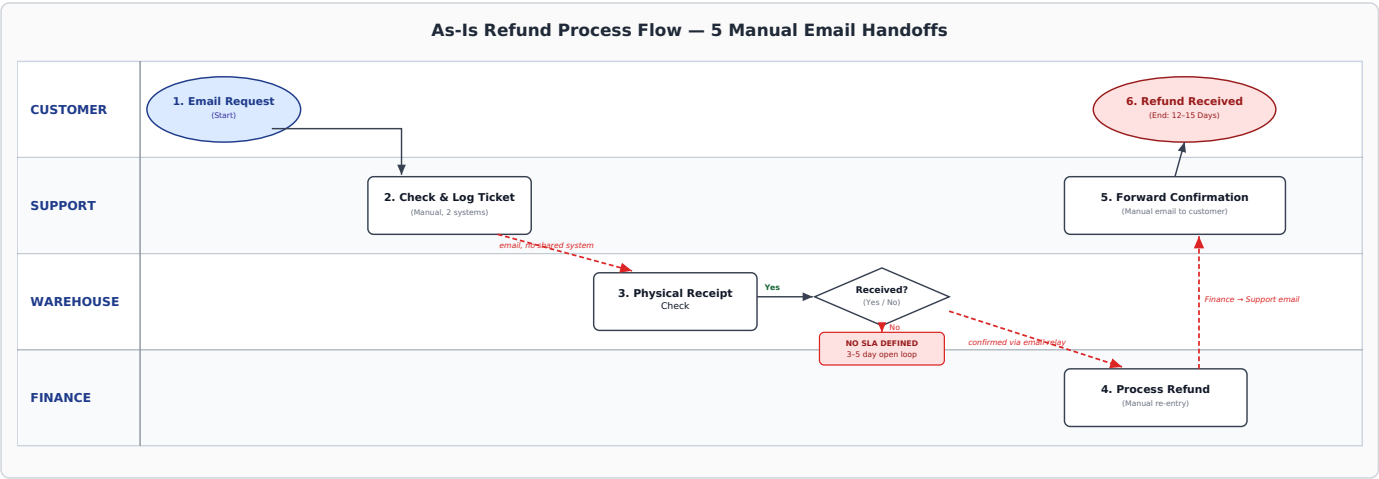
A mid-size Direct-to-Consumer (D2C) e-commerce brand operating with ~500 employees faced escalating customer churn and CSAT drops due to lengthy refund timelines. Customers regularly waited 12 to 15 business days to receive refund confirmations. Operational diagnostics revealed organizational silos: Support, Warehouse, and Finance operated in isolation, relying on unstructured, manual emails for cross-department handoffs. The single largest bottleneck occurred during return confirmation (3–5 days lost), where physical returns sat unconfirmed in the warehouse with no defined SLA or automated tracking trigger.

2. Stakeholder Analysis

Stakeholder	Interest	Influence	Impact / Key Concern
Leadership Sponsor (COO / VP Ops)	High	High	Owns customer churn and operational efficiency; funded project to achieve <5-day SLA.
Customer	High	Low (individually)	Directly experiences the 12–15 day wait; primary source of complaints and refund-related churn.
Finance Manager	Medium	Medium-High	Responsible for refund accuracy and audit control; cautious regarding automated payment approvals.
Support Team Lead & Agents	High	Medium / Low	Absorbed heavy manual workload chasing statuses; accountable for delays outside their control.
Warehouse Manager & Staff	Medium	Medium	Accountable for stock receipt confirmation; lacked shared visibility tools or a formal SLA.
IT / Systems Team	Low	Medium	Owns technical feasibility, system integrations, and API execution for the redesigned workflow.

3. Current State (As-Is) Process Flow

The As-Is process required 5 manual email handoffs across 3 disconnected systems (Order System, Ticketing System, and Payment System), causing queue delays and an open loop for unconfirmed returns.

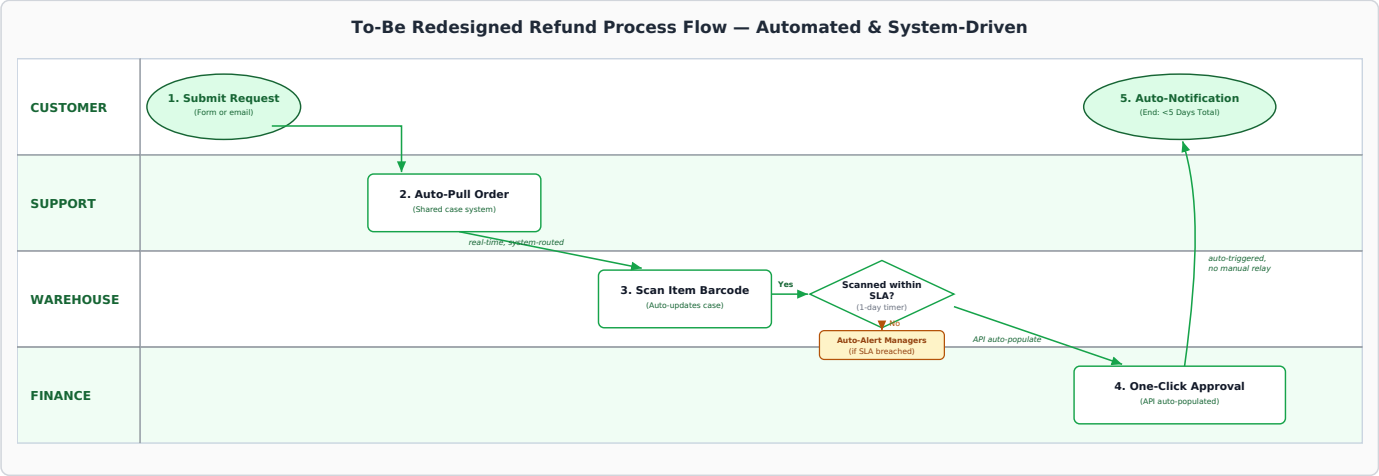


4. Gap Analysis & Time Breakdown

#	Identified Gap / Bottleneck	Root Cause	Time Impact
1	Unstructured Support-to-Warehouse Handoff	Escalations sent via manual email with no shared system tracking or receipt confirmation.	0.5 - 1 day
2	Lack of Warehouse Return Confirmation SLA	No defined timeframe or alert mechanism when returned packages sit unconfirmed.	3 - 5 days (Primary Bottleneck)
3	Manual Finance Re-entry	Finance system unintegrated; staff re-type order details from forwarded emails.	1 - 2 days
4	Manual Email Relay to Customer	No automated completion trigger; agents manually compose final confirmation emails.	0.5 - 1 day
5	Fragmented Systems (Order, Ticket, Payment)	Lack of unified integration/API layer forces staff to manually verify across 3 tools.	Systemic delay (~2-4 days backlog)

5. Future State (To-Be) Process Flow

The redesigned workflow establishes a shared case management system, automated barcode scan triggers, a 1-business-day Warehouse SLA with auto-escalation, and a Payment System API integration — eliminating every manual email handoff identified above.



6. Business Requirements Summary

Req ID	Functional Requirement Description	Priority
FR-1	System shall auto-pull order/customer details into a unified case record upon initiation.	Must Have
FR-2	System shall route case records directly to Warehouse queues without requiring email handoffs.	Must Have
FR-3	System shall update case status to "Return Confirmed" upon barcode scan in the warehouse.	Must Have
FR-4	System shall enforce a configurable 1-business-day SLA timer on Warehouse receipt verification.	Must Have
FR-5	System shall trigger automated alerts to Warehouse and Support managers upon SLA breach.	Must Have
FR-6	System shall auto-populate confirmed refund details into the Payment System via API.	Must Have
FR-7	System shall allow Finance to review and approve/reject each auto-populated refund case as a manual checkpoint, preserving financial control and audit compliance (approval is intentionally <i>not</i> automated).	Must Have
FR-8	System shall support automated customer email/SMS notifications immediately upon refund issuance.	Must Have

7. Cost-Benefit & ROI Analysis

Metric	Estimated Value	Basis / Assumptions
Implementation Cost	\$64,500 - \$97,500	Includes SaaS setup, scanner API development, payment API integration, and internal labor.
Support Labor Savings	~\$75,000 / year	Eliminates ~30 min/case of manual status chasing across ~500 refund cases/month (\$25/hr wage).
Finance Labor Savings	~\$45,000 / year	Saves ~15 min/case in re-keying payment data across ~500 cases/month (\$30/hr wage).
Churn Reduction Benefit	~\$35,000 - \$40,000 / year	10-15% reduction in refund-related churn (~\$150 LTV across monthly order base).
Total Annual Benefit	~\$155,000 - \$160,000	Net annual operational savings + revenue retention.
Projected Payback Period	5 - 8 Months	Ongoing annual benefit exceeds ~\$150k/year from Year 2 onward.

All figures above are stated, reasonable assumptions (case volume, wage rates, customer LTV) built for illustrative ROI purposes — not sourced from real company data.

8. User Acceptance Testing (UAT) Plan

Execution across a 20-case sample covering standard flows, high-value orders (>\$500), and intentional SLA breaches.

Key Test Scenarios: Validate order auto-pull (TC-1), auto-routing to Warehouse (TC-2), barcode scan status update (TC-3), SLA-breach manager alerts (TC-4), Finance API pre-population (TC-5), Finance approval/rejection handling (TC-6/TC-7), and automated customer notification (TC-8).

Sign-off Exit Criteria: 100% pass rate on Must-Have requirements; ≥90% (18/20) of test cases resolved in under 5 days; zero critical defects open.

9. Project Outcomes

65%+ Reduction in Refund Turnaround Time	0 Manual Cross-Department Handoffs (down from 5)	5-8 mo Projected Payback Period
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Key Performance Indicator	Baseline (As-Is)	Target Outcome (To-Be)
Average Refund Turnaround Time	12 - 15 Business Days	< 5 Business Days (>65% Reduction)
Manual Cross-Department Handoffs	5 Manual Emails	0 (100% System-Driven)
Warehouse Confirmation SLA	Untracked (3-5 Days)	1 Business Day (Enforced + Alerted)

10. Comprehensive Project Summary

This project transformed a legacy, high-friction customer refund process into a streamlined, automated workflow. By replacing unstructured email communication with a shared case system, barcode integration, and Payment System APIs, the initiative eliminated 5 manual handoffs and reduced processing cycle times by over 65%. The Finance approval step was deliberately preserved as a manual checkpoint rather than automated away, to protect financial control and audit compliance.

Core Strategic Takeaway: Operational bottlenecks in growing companies are rarely caused by employee effort — they are caused by structural visibility gaps and missing system integrations. Connecting isolated systems and establishing clear, system-enforced SLAs achieves dramatic operational efficiency without increasing departmental headcount.